

Question 1.**Marks**

- (a) Let $z = 5 - 2i$ and $w = 3 + 4i$.
- (i) Find $2w - \bar{w}$. 2
- (ii) Find $z(2w - \bar{w})$. 2
- (b) Given $\alpha = 1 - i\sqrt{3}$,
- (i) Find the exact value of $|\alpha|$ and $\text{Arg}(\alpha)$. 2
- (ii) Hence or otherwise, find the exact value of α^8 . 3
[express in the form $a + ib$, where a and b are real numbers]
- (c) Find the new complex number when $1 + i$ is rotated 45° anticlockwise about the origin O , in the Argand plane. 2
- (d) Find x and y , when $(x + iy)(2 + 3i) = 5 + 6i$. 2
- (e) Given $\psi^2 = 24 - 70i$, 2
- Find ψ in the form $a + ib$, where a and b are real.

Question 2.**[START A NEW PAGE]**

- (a) Shade the region of the Argand plane consisting of those variable points z for which:
- (i) $\text{Re}(z) < 2$ and $\text{Im}(z) > -1$. 1
- (ii) $|z - 1 - i| \leq 1$ and $0 \leq \arg z \leq \frac{\pi}{4}$. 2
- (b) Find the equation of the locus of z , if $|z - i| = \text{Im}(z)$. 3
Sketch the locus on an Argand plane.
- (c) Find the equation of the locus of z when $\arg\left(\frac{z-1}{z+1}\right) = \frac{\pi}{2}$. 2

Q 2 continued

- (d) Given that $z = x + iy$ is a variable point on the Argand plane such that $z\bar{z} - 2(z + \bar{z}) = 21$.
- (i) Find the locus of z . 2
- (ii) Hence determine the maximum value of: $|z - 4|$. 2
- (e) Given 1 , ω and ω^2 are the cube roots of unity and each are represented by the points A_1 , A_2 and A_3 respectively on an Argand plane. 3
- Find the value of $A_1A_2 \times A_1A_3$, where ' A_1A_2 ' represents the length A_1A_2 .

Question 3. [START A NEW PAGE]

- (a) Point A represents the complex number α and the point P represents the complex number z . 2
- Point P is rotated about the point A through a right angle in an anticlockwise direction to take up the new position, B , representing the complex number β .
- Find β (in terms of α and z).
- (b) Given $z = \cos \theta + i \sin \theta$,
- (i) Use De Moivre's theorem, to show that 2
- $$\cos n\theta = \frac{1}{2}(z^n + z^{-n})$$
- (ii) Hence deduce that: $\cos \theta \cos 2\theta = \frac{1}{2}(\cos \theta + \cos 3\theta)$ 3
- (c) Given the complex number z , such that $z = k(\cos \theta + i \sin \theta)$, 3
- where k is a real number and $0 < \theta < \pi$.
- Show that: $\arg(z + k) = \frac{1}{2}\theta$.

Q 3. continued

- (d) Given $z = x + iy$ is a variable point and $\alpha = a + ib$ is a fixed point on the Argand plane,
- (i) Show that $z\alpha - \bar{z}\bar{\alpha} = 0$ represents a straight line through the origin O . 2
- (ii) Suppose that z_1 and z_2 are the solutions to the simultaneous equations: $z\alpha - \bar{z}\bar{\alpha} = 0$ and $|z - \beta| = k$, where $\beta = p + iq$, and where p, q and k are positive real numbers, show that $|z_1||z_2| = |p^2 + q^2 - k^2|$. 3

Question 4. [START A NEW PAGE]

- (a) Let α, β and γ be the zeros of the polynomial function:

$$P(x) = x^3 + 2x^2 + 19x + 18.$$
- (i) Find $\alpha + \beta + \gamma$. 1
- (ii) Find $\alpha^2 + \beta^2 + \gamma^2$. 2
- (iii) Hence, or otherwise determine how many of the zeros are real. Give reasons. 2
- (b) Consider the polynomial equation: $z^5 - i = 0$,
- (i) Show that $z = i$ is a solution to the equation, and hence show that $1 - iz - z^2 + iz^3 + z^4 = 0$, for $z \neq i$. 2
- (ii) Hence or otherwise, find all the roots of $z^5 - i = 0$. [you may leave the roots in $\text{cis}\theta$ form] 2
- (iii) Show that $(z - i)[z^2 - 2i \sin \frac{\pi}{10} z - 1][z^2 + 2i \sin \frac{3\pi}{10} z - 1] = 0$. 4
- (iv) Hence show that: $\sin \frac{\pi}{10} \sin \frac{3\pi}{10} = \frac{1}{4}$. 2

THE END



Q1(a) $z = 5 - 2i$, $w = 3 + 4i$

(i) $2w - \bar{w} = 6 + 8i - (3 - 4i)$
 $= 3 + 12i$

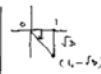
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(ii) $z(2w - \bar{w}) = (5 - 2i)(3 + 12i)$
 $= 15 + 60i - 6i + 24$
 $= 39 + 54i$

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(b) $\alpha = 1 - i\sqrt{3}$

(i) $|\alpha| = \sqrt{1^2 + (-\sqrt{3})^2}$
 $|\alpha| = 2$



$\arg \alpha = -\frac{\pi}{3}$

1 mark

(ii) $\alpha^8 = [2 \cos(-\frac{\pi}{3})]^8$
 $= 2^8 \cos(-\frac{8\pi}{3})$ (de Moivre's Thm)
 $= 256 \cos(-\frac{2\pi}{3})$
 $= 256 \cos \frac{4\pi}{3}$
 $= 256 (\cos \frac{4\pi}{3} + i \sin \frac{4\pi}{3})$
 $= 256 (-\frac{1}{2} - i\frac{\sqrt{3}}{2})$
 $\alpha^8 = -128 - 128i\sqrt{3}$

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(c) New complex no $= (1+i) \cos \frac{\pi}{4}$
 $= (1+i) \frac{1}{\sqrt{2}} (1+i)$
 $= \frac{2i\sqrt{2}}{2}$

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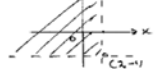
(d) $(x+iy)(2+3i) = 5+6i$
 $(x+iy) = \frac{(5+6i)(2-3i)}{(2+3i)(2-3i)}$
 $= \frac{10 - 15i + 12i + 18}{4+9}$
 $= \frac{28 - 3i}{13} = \frac{28}{13} - \frac{3}{13}i$

$\therefore \begin{cases} x = \frac{28}{13} \\ y = -\frac{3}{13} \end{cases}$

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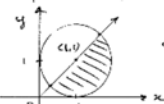
Q2(a)

(i) $\operatorname{Re}(z) < 3$ and $\operatorname{Im}(z) > -1$
 $x < 3$ and $y > -1$



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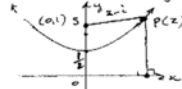
(ii) $|z - 1 - i| \leq 1$ and $0 \leq \arg z \leq \frac{\pi}{4}$
 $(x-1)^2 + (y-1)^2 \leq 1$
and $0 \leq \arg z \leq \frac{\pi}{4}$



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(b) Let $z = x+iy$
 $\operatorname{Im}(z) = y$, $|z-i| = \sqrt{x^2 + (y-1)^2}$
 $\therefore \sqrt{x^2 + (y-1)^2} = y$
For $y \geq 0$, $x^2 + (y-1)^2 = y^2$

$\therefore x^2 = 2y - 1 = 4(\frac{1}{2})(y - \frac{1}{2})$
A parabola
 $V = (0, \frac{1}{2})$



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(c) $\arg(\frac{z-1}{z+1}) = \frac{\pi}{2}$, let $z = x+iy$

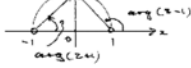
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Many methods: $x^2 + y^2 = 1$ for $|z| \leq 1$ and $\arg z < \frac{\pi}{2}$

• Geometric/Algebraic $y = \sqrt{1-x^2}$

• Trig. $\arg(z-1) = \frac{\pi}{2}$

• Algebraic $\arg(z+1)$



(e) $\sqrt{z} = 24 - 70i$

METHOD 1 $24 - 70i = 24 - 28i\sqrt{5}$
 $= 24 - 28i\sqrt{5}$
 $= (2-5i)^2$ or $(-7+5i)^2$
 $\therefore \sqrt{z} = 2-5i$ or $-7+5i$

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METHOD 2 Let $\sqrt{z} = x+iy$, $x, y \in \mathbb{R}$

$z = x^2 - y^2 + 2xyi$

$\therefore x^2 - y^2 = 24$ (1)

$2xy = -70$ (2)

$x^2y^2 = (-35)^2 = 1225$

$\therefore x^2 \times (1) x^4 - x^2y^2 = 24x^2$

i.e. $x^4 - 1225 = 24x^2$

$x^4 - 24x^2 - 1225 = 0$

$(x^2 - 49)(x^2 + 25) = 0$

$\therefore x^2 = 49$ or $x^2 = -25$

$\therefore x = \pm 7$ or no real roots pass.

So $y = \mp 5$ from (2)

$\therefore \sqrt{z} = 7-5i$ or $-7+5i$

METHOD 3 $\sqrt{z} = x+iy$

$|\sqrt{z}| = \sqrt{x^2 + y^2}$

$(\sqrt{z})^2 = |z| = x^2 + y^2$

and $|\sqrt{z}| = |24-70i| = \sqrt{24^2 + (-70)^2} = \sqrt{5476}$

i.e. $|\sqrt{z}| = 74$

$\therefore x^2 + y^2 = 74$ (1)

$x^2 - y^2 = 24$ (2)

$2x^2 = 98$

$x^2 = 49$

$x = \pm 7$

So $y^2 = 25 \Rightarrow y = \pm 5$

check now: $\sqrt{z} = 7-5i$ or $-7+5i$

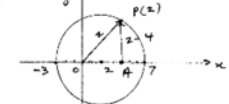
Q2(d) $z = 3(2+i) = 6+3i$

(i) let $z = x+iy$, $\bar{z} = x-iy$

$\therefore x^2 + y^2 - 2.2x = 21$

$\Rightarrow (x-2)^2 + y^2 = 25$

This represents a circle $C(2,0) r=5$



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(ii) Let $|z-4| = AP$ where $A = (4,0)$

max AP when P is at $(-3,0)$

$\therefore AP_{\max} = \max(|z-4|) = 3+4 = 7$

②

(c) METHOD 1 $z^3 = 1$, $|z| = 1$

$A_1A_2 = \omega - 1$, $A_1A_3 = \omega^2 - 1$

$A_2A_3 = \omega^2 - \omega$

$\therefore A_1A_2 \times A_1A_3 = (\omega - 1)(\omega^2 - 1)$

$= \omega^3 - \omega - \omega^2 + 1$

$= 1 - (\omega + \omega^2) + 1$

$= 1 - (-1) + 1$

$= 3$

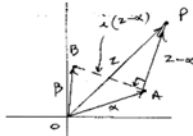
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METHOD 2. use cosine rule x2

3. Pythagoras with equil. Δ .

4.

Q3 (a)



$$\vec{AP} = z - x$$

rotate $\frac{\theta}{2}$ about A $\therefore \vec{AB} = \vec{z} - x$

$$\therefore \vec{OB} = \vec{p} = x + \vec{z} - x = (1-i)x + \vec{z}$$

(2)

(b) (i)

$$z = \cos \theta$$

$$z^n = (\cos \theta)^n = \cos n\theta \quad (\text{de Moivre})$$

ie $z^n = \cos n\theta + i \sin n\theta$

$$z^{-n} = (\cos \theta)^{-n} = \cos(-n\theta)$$

$$z^{-n} = \cos(-n\theta) + i \sin(-n\theta)$$

ie $z^{-n} = \cos n\theta - i \sin n\theta$ $\cos(-\theta) = \cos \theta$ $\sin(-\theta) = -\sin \theta$

$$\text{so } z^n + z^{-n} = 2 \cos n\theta$$

$$\therefore \cos n\theta = \frac{1}{2} (z^n + z^{-n})$$

(2)

(ii)

$$\text{now } \cos \theta = \frac{1}{2} (z + z^{-1})$$

$$\cos 2\theta = \frac{1}{2} (z^2 + z^{-2})$$

$$\therefore \cos \theta = \cos 2\theta$$

$$= \frac{1}{4} (z + z^{-1})(z^2 + z^{-2})$$

$$= \frac{1}{4} (z^3 + z^{-1} + z^{-1} + z^{-3})$$

$$= \frac{1}{4} [z + z^{-1} + z^3 + z^{-3}]$$

$$= \frac{1}{4} (2 \cos \theta + 2 \cos 3\theta)$$

$$= \frac{1}{2} (\cos \theta + \cos 3\theta)$$

(3)

Q3 (d) (i) $z = x + iy$, $\alpha = a + ib$

$$2x - \bar{z}\alpha = (x+iy)(a+ib) - (x-iy)(a-ib)$$

$$= ax - by + i(bx + ay) - [ax - by - i(bx + ay)]$$

$$= ax - by + i(bx + ay) - ax + by + i(bx + ay)$$

$$2x - \bar{z}\alpha = 2i(bx + ay)$$

$$\text{no } 2i(bx + ay) = 0$$

$$\Rightarrow bx + ay = 0 \quad \text{or } y = -\frac{b}{a}x$$

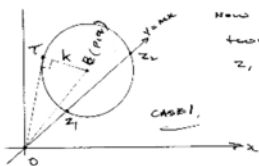
this represents a line thru 0

and it is of the form $ax + by + c = 0$, $c = 0$.

(2)

(ii) $|z - p| = k$ rep. a circle $B(p, q)$ $r = k$

$$2x - \bar{z}\alpha = 0 \text{ rep a line } y = mx$$

Now the line is a secant
two points of intersection
 z_1 and z_2 

$$\text{now } |z_1||z_2| = OP_1 \cdot OP_2 = OT^2 \quad (\text{sq. on tangent to circle equals product of intercepts on secant})$$

$$\text{so } |z_1||z_2| = OB^2 - r^2 \quad (\text{Rght. thm})$$

$$\therefore |z_1||z_2| = p^2 + q^2 - k^2$$

(3)

$$\text{or solve } (x-p)^2 + (y-q)^2 = k^2 \quad (1)$$

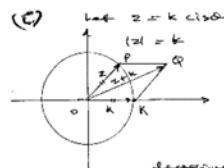
$$y = -\frac{b}{a}x \quad (2)$$

$$\text{to get } \left(\frac{a^2+b^2}{a^2+b^2}\right)x^2 + 2\left(\frac{bp}{a} - \frac{2p}{a}\right)x + p^2 + q^2 - k^2 = 0$$

$$\Rightarrow x_1 x_2 = \frac{p^2 + q^2 - k^2}{a^2 + b^2}$$

$$\text{then!!! get } |z_1||z_2| = \sqrt{x_1^2 + y_1^2} \sqrt{x_2^2 + y_2^2}$$

(c)



Arg $z = \theta$, $\vec{OK} = k$
Form parallelogram of vectors
 $\vec{OQ} = z + k$
 $OPQK$ is a rhombus
 OQ is a diagonal of the rhombus
diagonal of rhombus bisects interior $\angle KOP$
 $\therefore \angle KOP = \arg(z+k) = \frac{1}{2}\theta$

or Algebraic: $z+k = k(1 + \cos \theta)$
 $= k(1 + \cos \theta + i \sin \theta)$

$$= k(2 \cos^2 \frac{\theta}{2} + i 2 \sin \frac{\theta}{2} \cos \frac{\theta}{2})$$

$$0 \leq \theta < \pi$$

$$z+k = 2k \cos \frac{\theta}{2} \left[\cos \frac{\theta}{2} + i \sin \frac{\theta}{2} \right]$$

$$\therefore \arg(z+k) = 0 + \frac{1}{2}\theta$$

$$= \frac{1}{2}\theta$$

(3)



Q4

$$P(x) = x^3 + 2x^2 + 4x + 16$$

$$(a) (i) x + p + q = -\frac{b}{a} = -2 \quad (1)$$

$$(ii) x^2 + p^2 + q^2 = (x+p+q)^2 - 2(xp + pq + xq)$$

$$= (-2)^2 - 2 \times 19$$

$$= -34 \quad (2)$$

(iii) 1 real zero because:

• If all the zeros were real $\sum x_i^2 > 0$

$$\text{but } \sum x_i^2 < 0$$

• there is at least one complex zero

and as the coeffs of $P(x)$ are all real

• complex zeros are in conjugate pairs

and as $\deg P(x) = 3$, by F.T.Hthere are $3 - 2 = 1$ real zero• Solving $P(x) = 0$

$$\text{get } P(-1) = 0$$

$$\therefore (x+1)(x^2 + x + 16) = 0$$

$$\text{for } x^2 + x + 16 \quad \Delta = 1 - 64 < 0$$

• 2 unreal roots

• only one real zero ($x = -1$)

(2)

(b) (i)

$$z^5 - i = z^5 - i$$

$$= x^5 i - i$$

$$= (x^5 - 1)i$$

$$= 0$$

• $z = i$ is a solution

$$\text{to show } (-i - z^2 + iz^3 + z^4 = 0)$$

$$\text{use long division } z^5 - i \div z^2 - z^2 \dots - i$$

$$\text{or geometric series } 1 + (-i) + (-i)^2 + \dots + (-i)^{n-1} = \frac{1 - (-i)^n}{1 - (-i)}$$

$$\text{or show } (z^2)(1 - i - z^2 + iz^3 - z^4) = z^5 - i = 0 \text{ by expanding.}$$

$$\therefore 1 - i - z^2 + iz^3 - z^4 = 0 \text{ so } z = i \neq 0.$$

$$1 + (-i) + (-i)^2 + \dots + (-i)^{n-1} = \frac{1 - (-i)^n}{1 - (-i)}$$

Q4(b) (ii) $z^5 - i = 0$

Since one root is $i = e^{i\frac{\pi}{2}}$
and the roots are equally spaced
around the circle by $\frac{2\pi}{5}$, $n=5$.

$z_1 = i = e^{i\frac{\pi}{2}}$

And $z_2 = e^{i(\frac{\pi}{2} + \frac{2\pi}{5})} = e^{i\frac{9\pi}{10}} = \cos\frac{9\pi}{10} + i\sin\frac{9\pi}{10}$

$z_3 = e^{i(\frac{\pi}{2} + \frac{4\pi}{5})} = e^{i\frac{13\pi}{10}} = \cos\frac{13\pi}{10} + i\sin\frac{13\pi}{10}$

$z_4 = e^{i(\frac{\pi}{2} + \frac{6\pi}{5})} = e^{i\frac{17\pi}{10}} = \cos\frac{17\pi}{10} + i\sin\frac{17\pi}{10}$

$z_5 = e^{i(\frac{\pi}{2} + \frac{8\pi}{5})} = e^{i\frac{21\pi}{10}} = \cos\frac{21\pi}{10} + i\sin\frac{21\pi}{10}$



or let $z = re^{i\theta}$

$z^5 = r^5 e^{i5\theta} = i = (e^{i\frac{\pi}{2}})^5, k=0,1,2,3,4$
generated above roots.

ii) As $z^5 - i = (z - z_1)(z - z_2)(z - z_3)(z - z_4)(z - z_5) = 0$

i.e. $(z - i)(z - z_2)(z - z_3)(z - z_4)(z - z_5)$

and noting:

$(z - \alpha)(z - \beta) = z^2 - (\alpha + \beta)z + \alpha\beta$

With z_1 and z_5 , z_2 and z_4

and eq: $\alpha\beta = (-\cos\frac{9\pi}{10} + i\sin\frac{9\pi}{10})(\cos\frac{17\pi}{10} + i\sin\frac{17\pi}{10})$
 $= -(\cos\frac{9\pi}{10} - i\sin\frac{9\pi}{10})(\cos\frac{17\pi}{10} + i\sin\frac{17\pi}{10})$
 $= -1(\cos\frac{26\pi}{10} + i\sin\frac{26\pi}{10})$
 $= -1$

so $z^5 - i = (z - i)[z^2 - (z_2 + z_4)z + 1][z^2 - (z_3 + z_5)z + 1] = 0$
i.e. $(z - i)(z^2 - 2i\sin\frac{9\pi}{10}z + 1)(z^2 + 2i\sin\frac{9\pi}{10}z + 1) = 0$ of mod.

iv) As $(z - i)[z^2 - 2i\sin\frac{9\pi}{10}z + 1][z^2 + 2i\sin\frac{9\pi}{10}z + 1] = z^5 - i$

i.e. $(z - i)(1 - iz - z^2 + iz^3 - z^4) = z^5 - i$

the coeff. of z^2 in $[1 - iz - z^2 + iz^3 - z^4]$

is $-1 - 4i^2\sin\frac{9\pi}{10}\sin\frac{9\pi}{10} = -1$

i.e. $-2 + 4\sin\frac{9\pi}{10}\sin\frac{9\pi}{10}$

but coeff of z^2 in $1 - iz - z^2 + iz^3 - z^4$ is -1

so $-2 + 4\sin\frac{9\pi}{10}\sin\frac{9\pi}{10} = -1$

$\Rightarrow \sin\frac{9\pi}{10} \times \sin\frac{9\pi}{10} = \frac{1}{4}$ QED

(2)

(4)